

trgn515_final_assignment

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Task 1: Bioinformatic pipeline

1. List the files of the input FastQ folder. Of the sequencing technologies reviewed in this course, what's the most likely technology used for this dataset? Why? (5pts)

```
[ranzhu@discovery1 final_project]$ ls -lh /project2/msalomon_1816/trgn_515/8_final/fastq/
total 169M
-rw-rw-r-- 1 atortiz msalomon_1816 84M May  3 16:07 a549_1.fq.gz
-rw-rw-r-- 1 atortiz msalomon_1816 85M May  3 16:07 a549_2.fq.gz
[ranzhu@discovery1 final_project]$
```

- The FastQ folder contains two files, a549_1.fq.gz and a549_2.fq.gz, indicating paired-end sequencing data where each DNA fragment is sequenced from both ends. The presence of _1 and _2 files with nearly identical sizes (84MB and 85MB) is characteristic of Illumina paired-end sequencing, where both reads have the same length. Additionally, the reference genome index provided is a BWA index, which is specifically designed for aligning short Illumina reads (~100-150bp) to a reference genome. Thus, this dataset was generated using Illumina.

2. Outline the steps of your suggested pipeline to identify and annotate SNPs and SVs of clinical significance and name the bioinformatic software used for each step (15pts)

Step	Description	Input	Output	Software
1	Pre-trimming QC	fastq.gz	QC report	FastQC
2	Trimming	fastq.gz	trimmed fastq.gz.	fastp
3	Post-trimming QC	trimmed fastq.gz	QC report	FastQC
4	Align reads to GRCh38	trimmed fastq.gz	BAM	BWA-MEM
5	Sort and mark duplicates	BAM	sorted + dedup BAM	Samtools markup
6	SNP & Indel variant calling	BAM	VCF	bcftools mpileup + bcftools call
7	SV calling	BAM	SV VCF	delly
8	Filter SNPs/ Indels by quality	VCF	filtered VCF	bcftools filter

Step	Description	Input	Output	Software
9	Filter SV by quality	SV_VCF	filtered SV_VCF	bcftools filter
10	VCF SNPs/Indels Annotation clinical info	filtered VCF	annotated VCF	bcftools annotate + GnomAD + ClinVar
11	VCF SVs Annotation - functional	filtered SV_VCF	annotated SV_VCF	VEP

3. Attach a SLURM job with the entire pipeline (10pts)

```
#!/bin/bash
#SBATCH --account=msalomon_1816
#SBATCH --partition=main
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --mem=64G
#SBATCH --time=12:00:00
#SBATCH --job-name=A549_pipeline
#SBATCH --output=/project2/msalomon_1816/ranzhu/final_project/logs/pipeline_%j.out
#SBATCH --error=/project2/msalomon_1816/ranzhu/final_project/logs/pipeline_%j.err

source ~/.bashrc
conda activate /project2/msalomon_1816/trgn_515/conda_env/trgn515

SAMPLE=a549
in_dir=/project2/msalomon_1816/trgn_515/8_final/fastq
fastqc_dir=/project2/msalomon_1816/ranzhu/final_project/fastqc
fastp_dir=/project2/msalomon_1816/ranzhu/final_project/fastp
aln_dir=/project2/msalomon_1816/ranzhu/final_project/alignments
vcf_dir=/project2/msalomon_1816/ranzhu/final_project/vcf
sv_dir=/project2/msalomon_1816/ranzhu/final_project/sv
ann_dir=/project2/msalomon_1816/ranzhu/final_project/annotated

REF_BWA=/project2/biodb/genomes/Homo_sapiens/NCBI/GRCh38/Sequence/BWAIndex/genome.fa
REF_VC=/project2/biodb/genomes/Homo_sapiens/NCBI/GRCh38/Sequence/WholeGenomeFasta/genome.fa
GNOMAD=/project2/msalomon_1816/trgn_515/8_final/db/gnomad.regions.vcf.gz
CLINVAR=/project2/msalomon_1816/trgn_515/8_final/db/clinvar.vcf.gz
VEP_CACHE=/project2/msalomon_1816/trgn_515/8_final/db/vep_cache

mkdir -p $fastqc_dir/pre_trimming \
        $fastqc_dir/post_trimming \
        $fastp_dir \
        $aln_dir \
        $vcf_dir \
        $sv_dir \
        $ann_dir \
        /project2/msalomon_1816/ranzhu/final_project/logs

# === STEP 0: Pre-trimming FastQC =====
echo "[$(date)] Step 0: Pre-trimming FastQC"
```

```

fastqc \
  -o $fastqc_dir/pre_trimming \
  --extract \
  --thread 4 \
  $in_dir/${SAMPLE}_1.fq.gz \
  $in_dir/${SAMPLE}_2.fq.gz

# === STEP 1: Trimming with fastp ===
echo "[$(date)] Step 1: Trimming with fastp"

fastp \
  -i $in_dir/${SAMPLE}_1.fq.gz \
  -I $in_dir/${SAMPLE}_2.fq.gz \
  -o $fastp_dir/${SAMPLE}_1.trimmed.fq.gz \
  -O $fastp_dir/${SAMPLE}_2.trimmed.fq.gz \
  --thread 4 \
  --json $fastp_dir/${SAMPLE}.fastp.json \
  --html $fastp_dir/${SAMPLE}.fastp.html

# === STEP 2: Post-trimming FastQC ===
echo "[$(date)] Step 2: Post-trimming FastQC"

fastqc \
  -o $fastqc_dir/post_trimming \
  --extract \
  --thread 4 \
  $fastp_dir/${SAMPLE}_1.trimmed.fq.gz \
  $fastp_dir/${SAMPLE}_2.trimmed.fq.gz

# === STEP 3: Alignment with BWA-MEM ===
echo "[$(date)] Step 3: Aligning reads with BWA-MEM"

flowcell=$(zcat $fastp_dir/${SAMPLE}_1.trimmed.fq.gz | head -n1 | cut -d':' -f3)
lane=$(zcat $fastp_dir/${SAMPLE}_1.trimmed.fq.gz | head -n1 | cut -d':' -f4)
rg_id=${flowcell}.${lane}
pu=${flowcell}.${lane}.${SAMPLE}

bwa mem -t 4 \
  -R "@RG\tID:$rg_id\tPU:$pu\tPL:ILLUMINA\tSM:${SAMPLE}" \
  $REF_BWA \
  $fastp_dir/${SAMPLE}_1.trimmed.fq.gz \
  $fastp_dir/${SAMPLE}_2.trimmed.fq.gz \
  | samtools view -bS - \
  | samtools sort \
  -@ 4 \
  -T $aln_dir \
  -O bam \
  -o $aln_dir/${SAMPLE}.bam

# === STEP 4: Mark Duplicates and Index ===
echo "[$(date)] Step 4: Marking duplicates"

samtools collate -Ou $aln_dir/${SAMPLE}.bam | \

```

```

samtools fixmate -m - - -u | \
samtools sort - -u | \
samtools markdup - $aln_dir/${SAMPLE}.dedup.bam

samtools index $aln_dir/${SAMPLE}.dedup.bam

# === STEP 5: SNP & Indel Calling =====
echo "[$(date)] Step 5: SNP/Indel calling with bcftools"

bcftools mpileup \
  -f ${REF_VC} \
  -a AD,INFO/AD,ADF,INFO/ADF,ADR,INFO/ADR,DP \
  -Q 20 \
  -r chr9,chr12 \
  -Ou \
  $aln_dir/${SAMPLE}.dedup.bam \
  | bcftools call \
  -m \
  -v \
  -Oz -o $vcf_dir/${SAMPLE}.genotyped.vcf.gz

tabix -f -p vcf $vcf_dir/${SAMPLE}.genotyped.vcf.gz

# === STEP 6: SV Calling with Delly =====
echo "[$(date)] Step 6: SV calling with Delly"

delly call \
  -g ${REF_VC} \
  -o ${sv_dir}/${SAMPLE}_sv.bcf \
  $aln_dir/${SAMPLE}.dedup.bam

# Convert BCF to VCF.gz
bcftools view ${sv_dir}/${SAMPLE}_sv.bcf \
  -Oz -o ${sv_dir}/${SAMPLE}_sv.vcf.gz

tabix -f -p vcf ${sv_dir}/${SAMPLE}_sv.vcf.gz

# Filter to chr9 and chr12 only
bcftools view -r chr9,chr12 \
  ${sv_dir}/${SAMPLE}_sv.vcf.gz \
  -Oz -o ${sv_dir}/${SAMPLE}.sv.chr9_12.vcf.gz

tabix -f -p vcf ${sv_dir}/${SAMPLE}.sv.chr9_12.vcf.gz

# === STEP 7: Filter SNPs and Indels =====
echo "[$(date)] Step 7: Filtering SNPs and Indels"

# Filter SNPs
bcftools view -v snps \
  $vcf_dir/${SAMPLE}.genotyped.vcf.gz | \
  bcftools filter -m+ -s 'LowQUAL' -e 'QUAL < 20' | \
  bcftools filter -m+ -s 'LowMQ' -e 'MQ < 20' | \
  bcftools filter -m+ -s 'LowDP' -e 'INFO/DP < 20' | \

```

```

bcftools filter -m+ -s 'LowAD' -e 'FMT/AD[:1] < 10' \
-Oz -o $vcf_dir/${SAMPLE}.snps.filtered.vcf.gz

tabix -f -p vcf $vcf_dir/${SAMPLE}.snps.filtered.vcf.gz

# Filter Indels
bcftools view -v indels \
$vcf_dir/${SAMPLE}.genotyped.vcf.gz | \
bcftools filter -m+ -s 'LowQUAL' -e 'QUAL < 20' | \
bcftools filter -m+ -s 'LowMQ' -e 'MQ < 20' | \
bcftools filter -m+ -s 'LowDP' -e 'INFO/DP < 20' | \
bcftools filter -m+ -s 'LowAD' -e 'FMT/AD[:1] < 10' \
-Oz -o $vcf_dir/${SAMPLE}.indels.filtered.vcf.gz

tabix -f -p vcf $vcf_dir/${SAMPLE}.indels.filtered.vcf.gz

# === STEP 8: Filter SVs =====
echo "[$(date)] Step 8: Filtering SVs"

bcftools filter \
-i 'INFO/MAPQ >= 20 && INFO/PE >= 10 && INFO/SR >= 10 && FMT/DV >= 20' \
${sv_dir}/${SAMPLE}.sv.chr9_12.vcf.gz \
-Oz -o ${sv_dir}/${SAMPLE}.sv.hq.vcf.gz

tabix -f -p vcf ${sv_dir}/${SAMPLE}.sv.hq.vcf.gz

# === STEP 9: Annotate SNPs/Indels with GnomAD and ClinVar =====
echo "[$(date)] Step 9: Annotating SNPs/Indels with GnomAD and ClinVar"

# Annotate SNPs
bcftools annotate \
-a ${GNOMAD} \
-c ID,INFO/AF \
-Oz -o ${ann_dir}/${SAMPLE}.snps.gnomad.vcf.gz \
$vcf_dir/${SAMPLE}.snps.filtered.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.snps.gnomad.vcf.gz

bcftools annotate \
-a ${CLINVAR} \
-c INFO/CLNSIG \
-Oz -o ${ann_dir}/${SAMPLE}.snps.gnomad.clinvar.vcf.gz \
${ann_dir}/${SAMPLE}.snps.gnomad.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.snps.gnomad.clinvar.vcf.gz

# Annotate Indels
bcftools annotate \
-a ${GNOMAD} \
-c ID,INFO/AF \
-Oz -o ${ann_dir}/${SAMPLE}.indels.gnomad.vcf.gz \
$vcf_dir/${SAMPLE}.indels.filtered.vcf.gz

```

```

tabix -f -p vcf ${ann_dir}/${SAMPLE}.indels.gnomad.vcf.gz

bcftools annotate \
  -a ${CLINVAR} \
  -c INFO/CLNSIG \
  -Oz -o ${ann_dir}/${SAMPLE}.indels.gnomad.clinvar.vcf.gz \
  ${ann_dir}/${SAMPLE}.indels.gnomad.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.indels.gnomad.clinvar.vcf.gz

# === STEP 10: Annotate SVs with VEP =====
echo "[$(date)] Step 10: Annotating SVs with VEP"

conda deactivate
conda activate /project2/msalomon_1816/trgn_515/conda_env/vep

vep \
  -i ${sv_dir}/${SAMPLE}.sv.hq.vcf.gz \
  -o ${ann_dir}/${SAMPLE}.sv.vep.vcf.gz \
  --vcf --compress_output bgzip \
  --species homo_sapiens --assembly GRCh38 \
  --offline --cache \
  --dir_cache ${VEP_CACHE} \
  --symbol --canonical \
  --sift b --polyphen b \
  --fork 4

tabix -f -p vcf ${ann_dir}/${SAMPLE}.sv.vep.vcf.gz

echo "[$(date)] Pipeline complete!"
echo "Results are in: ${ann_dir}/"

```

Task 2: SNPs and Indels

1. Paste your filtering command (5pts)

- SNP filtering command

```

bcftools view -v snps \
  $vcf_dir/${SAMPLE}.genotyped.vcf.gz | \
  bcftools filter -m+ -s 'LowQUAL' -e 'QUAL < 20' | \
  bcftools filter -m+ -s 'LowMQ' -e 'MQ < 20' | \
  bcftools filter -m+ -s 'LowDP' -e 'INFO/DP < 20' | \
  bcftools filter -m+ -s 'LowAD' -e 'FMT/AD[:1] < 10' \
  -Oz -o $vcf_dir/${SAMPLE}.snps.filtered.vcf.gz

```

- Indel filtering command:

```

bcftools view -v indels \
  $vcf_dir/${SAMPLE}.genotyped.vcf.gz | \

```

```
bcftools filter -m+ -s 'LowQUAL' -e 'QUAL < 20' | \
bcftools filter -m+ -s 'LowMQ' -e 'MQ < 20' | \
bcftools filter -m+ -s 'LowDP' -e 'INFO/DP < 20' | \
bcftools filter -m+ -s 'LowAD' -e 'FMT/AD[:1] < 10' \
-Oz -o $vcf_dir/${SAMPLE}.indels.filtered.vcf.gz
```

2. How many high confident SNPs did you identify? How many high confident Indels? (5pts)

```
# Count high confidence SNPs
bcftools view -f PASS \
/project2/msalomon_1816/ranzhu/final_project/vcf/a549.snps.filtered.vcf.gz \
| grep -v "^#" | wc -l

# Count high confidence Indels
bcftools view -f PASS \
/project2/msalomon_1816/ranzhu/final_project/vcf/a549.indels.filtered.vcf.gz \
| grep -v "^#" | wc -l
```

- Using the filtering criteria (QUAL $\geq$ 20, MQ $\geq$ 20, DP $\geq$ 20, AD $\geq$ 10), 4,697 high confidence SNPs and 507 high confidence Indels were identified in chromosomes 9 and 12 of the A549 lung adenocarcinoma cell line.

3. Are all these variants causing the cancer? Explain your answer (5pts)

- The 4697 SNPs and 507 Indels were identified by comparing the A549 cell line against the GRCh38 human reference genome. This means the identified variants include both germline variants and somatic variants. However, I cannot distinguish germline variants from somatic variants based on the variant call alone.
- Also, somatic variants can drive cancer, as they are mutations acquired in cancer cells that can affect oncogenes or tumor suppressor genes. Germline variants are inherited from parents and are present in all cells of the body as common population polymorphisms, and are generally not cancer-causing.
- Therefore, further annotation using GnomAD and ClinVar is needed. GnomAD provides population allele frequencies to filter out common germline variants, while ClinVar provides clinical significance annotations to identify pathogenic variants. Only after this clinical annotation, I can identify which variants are truly driving the cancer in A549.

4. Annotate your SNPs and Indels using the GnomAD and ClinVar databases. Suggest filters using the information from these databases. Why did you choose those filters? (10pts)

- SNPs and Indels were annotated using bcftools annotate with GnomAD and ClinVar databases:

```
# Annotate SNPs
bcftools annotate \
-a ${GNOMAD} \
-c ID,INFO/AF \
-Oz -o ${ann_dir}/${SAMPLE}.snps.gnomad.vcf.gz \
```

```

$vcf_dir/${SAMPLE}.snps.filtered.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.snps.gnomad.vcf.gz

bcftools annotate \
-a ${CLINVAR} \
-c INFO/CLNSIG \
-Oz -o ${ann_dir}/${SAMPLE}.snps.gnomad.clinvar.vcf.gz \
${ann_dir}/${SAMPLE}.snps.gnomad.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.snps.gnomad.clinvar.vcf.gz

# Annotate Indels
bcftools annotate \
-a ${GNOMAD} \
-c ID,INFO/AF \
-Oz -o ${ann_dir}/${SAMPLE}.indels.gnomad.vcf.gz \
$vcf_dir/${SAMPLE}.indels.filtered.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.indels.gnomad.vcf.gz

bcftools annotate \
-a ${CLINVAR} \
-c INFO/CLNSIG \
-Oz -o ${ann_dir}/${SAMPLE}.indels.gnomad.clinvar.vcf.gz \
${ann_dir}/${SAMPLE}.indels.gnomad.vcf.gz

tabix -f -p vcf ${ann_dir}/${SAMPLE}.indels.gnomad.clinvar.vcf.gz

```

- After annotation, I examined the allele frequency distribution of the annotated variants.

```

# Count variants with AF annotation
bcftools query -f '%AF\n' \
/project2/msalomon_1816/ranzhu/final_project/annotated/a549.snps.gnomad.clinvar.vcf.gz \
| grep -v "^\. $" | wc -l

# Count variants with AF > 0.01
bcftools query -f '%AF\n' \
/project2/msalomon_1816/ranzhu/final_project/annotated/a549.snps.gnomad.clinvar.vcf.gz \
| grep -v "^\. $" | awk '$1 > 0.01' | wc -l

```

- Results shows:
- Variants with AF annotation count about 6,117
- Variants count with AF > 0.01 about 5,990
- Variants with AF < 0.01 counts 127
- Variants with no AF annotation counts 580.
- Out of 6,117 variants with AF annotation, 5,990 (97.9%) have AF > 0.01, meaning the vast majority are common population polymorphisms present in healthy individuals and are unlikely to be cancer-driving mutations. Therefore, I chose AF < 0.01 as the filter threshold to remove these common

germline variants, which is a widely accepted cutoff in population genetics to distinguish common variants from rare disease-associated variants. Variants with no AF annotation (AF = ".") are also kept, as they may represent novel variants not yet catalogued in GnomAD.

- Then, I checked the ClinVar clinical significance annotations:

```
bcftools query -f '%CLNSIG\n' \
/project2/msalomon_1816/ranzhu/final_project/annotated/a549.snps.gnomad.clinvar.vcf.gz \
| sort | uniq -c | sort -rn | head -20
```

- The results show that only 1 variant is annotated as Pathogenic, while the majority have no ClinVar annotation (6,214) or are Benign/Likely_benign (16/7).
- Therefore, suggested filters set up:

Filter 1: Remove common population variants using GnomAD AF

```
bcftools filter \
-i 'INFO/AF < 0.01 || INFO/AF == "."' \
${ann_dir}/${SAMPLE}.snps.gnomad.clinvar.vcf.gz \
-Oz -o ${ann_dir}/${SAMPLE}.snps.rare.vcf.gz
```

Filter 2: Keep only Pathogenic variants using ClinVar

```
bcftools filter \
-i 'INFO/CLNSIG="Pathogenic" || INFO/CLNSIG="Likely_pathogenic"' \
${ann_dir}/${SAMPLE}.snps.gnomad.clinvar.vcf.gz \
-Oz -o ${ann_dir}/${SAMPLE}.snps.pathogenic.vcf.gz
```

- The GnomAD AF < 0.01 filter was chosen because the majority of variants (5,990 out of 6,117) have an allele frequency greater than 1% in the general population, meaning they are common germline polymorphisms that are unlikely to be cancer-causing. By filtering out these common variants and keeping only rare variants (AF < 0.01) or variants with no AF annotation, I can focus on variants that are more likely to be somatic mutations driving cancer.
- The ClinVar Pathogenic filter was chosen because I want to prioritize variants that are already known to be disease-causing based on existing clinical evidence. From my result, only 1 variant is annotated as Pathogenic, while the majority are either Benign, Likely_benign, or have no annotation. By keeping only Pathogenic or Likely_pathogenic variants, I can identify variants with known clinical significance, while excluding Benign and Likely_benign variants that are not expected to drive cancer.

5. Given your annotations and filters, what are the SNPs/Indels driving this cancer? Search for this position in ClinVar. Report the Gene Name and the genomic coordinates. (10pts)

Search for Pathogenic SNPs

```
bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\t%AF\t%CLNSIG\n' \
/project2/msalomon_1816/ranzhu/final_project/annotated/a549.snps.gnomad.clinvar.vcf.gz \
| grep "Pathogenic"
```

Search for Pathogenic Indels

```
bcftools query -f '%CHROM\t%POS\t%REF\t%ALT\t%AF\t%CLNSIG\n' \
/project2/msalomon_1816/ranzhu/final_project/annotated/a549.indels.gnomad.clinvar.vcf.gz \
| grep "Pathogenic"
```

- The results showed that only one Pathogenic SNP was identified at chr12:25245351 (C→T), while no Pathogenic Indels were found.
- Searching using GRCh38:chr12:25245351 position in ClinVar. KRAS is located on the reverse strand, the C→T change on the forward strand corresponds to a G→A change on the reverse strand. Therefore the confirmed variant is NM_004985.5(KRAS):c.34G>A (p.Gly12Ser), which is KRAS G12S.
- Gene Name: KRAS
- Genomic coordinates (GRCh38): chr12:25245351
- Cytogenetic location: 12p12.1
- Variant: NM_004985.5(KRAS):c.34G>A (p.Gly12Ser) — KRAS G12S
- Type: Single nucleotide variant (missense variant)
- Classification: Pathogenic (Germline), Oncogenic (Somatic)
- The KRAS G12S mutation (chr12:25245351, C→T) was identified as the only Pathogenic variant in my analysis. KRAS is an oncogene that acts as a molecular switch regulating cell growth. The G12S missense mutation occurs at codon 12, which has been reported in more than 2,000 cases in the COSMIC cancer database (COSMIC ID: COSV55497461), indicating that this is a recurrent mutation site in cancer. According to ClinVar, functional studies have shown that alterations at codon 12 result in downstream MAPK/ERK pathway activation and tumor formation (Park JT, et al., PMID: 25065594), leading to uncontrolled cell proliferation. According to ClinVar, germline variant is classified as pathogenic that is associated with Non-small cell lung carcinoma, Gastric cancer, RASopathy. Somatic variant is associated with Tier I (Strong) diagnostic for Rhabdomyosarcoma, Tier II (Potential) for Neuroblastoma, and Oncogenic for Neoplasm. The association of this variant with Non-small cell lung carcinoma is consistent with A549 being a lung adenocarcinoma cell line, suggesting that the KRAS G12S mutation is likely driving cancer in this cell line.

Task 3: SVs

1. Paste your filtering command (5pts)

```
bcftools filter \
-i 'INFO/MAPQ >= 20 && INFO/PE >= 10 && INFO/SR >= 10 && FMT/DV >= 20' \
/project2/msalomon_1816/ranzhu/final_project/sv/a549.sv.chr9_12.vcf.gz \
-Oz -o /project2/msalomon_1816/ranzhu/final_project/sv/a549.sv.hq.vcf.gz

bcftools view /project2/msalomon_1816/ranzhu/final_project/sv/a549.sv.hq.vcf.gz | grep -v "^#" | wc -l
```

2. Annotate your high confident SVs using VEP. Extract the name of the gene where the SV falls in. Find information about the identified genes. Which gene or genes are most likely to be linked to cancer? (10pts)

- Annotate high confident SVs using VEP.

```
/project2/msalomon_1816/trgn_515/conda_env/vep/bin/perl \
/project2/msalomon_1816/trgn_515/conda_env/vep/bin/vep \
-i /project2/msalomon_1816/ranzhu/final_project/sv/a549.sv.hq.vcf.gz \
-o /project2/msalomon_1816/ranzhu/final_project/annotated/a549.sv.vep.vcf.gz \
--vcf --compress_output bgzip \
--species homo_sapiens --assembly GRCh38 \
--offline --cache \
--dir_cache /project2/msalomon_1816/trgn_515/8_final/db/vep_cache \
```

```
--symbol --canonical \
--sift b --polyphen b \
--fork 4
```

- Extract the name of the gene where the SV falls in. Information about the identified genes:

SV	Position	Type	Genes
SV1	chr9:21832461	INV	MTAP
SV2	chr9:21907128	INV	CDKN2A, CDKN2B, MTAP, (UBA52P6 pseudogene)
SV3	chr9:22999488	DEL	LINC03142
SV4	chr9:23362803	DEL	LINC03142
SV5	chr12:26358087	DEL	ITPR2

```
zcat /project2/msalomon_1816/ranzhu/final_project/annotated/a549.sv.vep.vcf.gz | \
grep -v "^#" | \
grep -E "21832461|21907128|22999488|23362803|26358087" | \
grep -oP '\|[A-Z][A-Z0-9]+\|ENSG' | \
sed 's/|//g' | sed 's/ENSG//g' | \
sort -u
```

- Results: CDKN2A, CDKN2B, ITPR2, LINC03142, MTAP,(UBA52P6 pseudogene)
- UBA52P6 is a pseudogene and does not encode a functional protein, so it is not considered further in this analysis.
- Based on the VEP annotation results and gene information retrieved from the NCBI Gene database:
 1. CDKN2A (cyclin dependent kinase inhibitor 2A, GeneID: 1029) is a protein-coding tumor suppressor gene located on chromosome 9p21.3. According to NCBI, it encodes two important proteins: p16INK4a, which inhibits CDK4 kinase and prevents cell cycle progression, and p14ARF, which stabilizes the tumor suppressor protein p53 by sequestering MDM2. Through the regulatory roles of CDK4 and p53 in cell cycle G1 progression, both isoforms share a common functionality in cell cycle G1 control. This gene is frequently mutated or deleted in a wide variety of tumors and is known to be an important tumor suppressor gene. In my VEP results, the inversion at chr9:21907128 causes a coding_transcript_variant in CDKN2A.
 2. CDKN2B (cyclin dependent kinase inhibitor 2B, GeneID: 1030) is a protein-coding tumor suppressor gene that lies adjacent to CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. According to NCBI, this gene encodes a cyclin-dependent kinase inhibitor (p15INK4b) which forms a complex with CDK4 or CDK6 and prevents the activation of the CDK kinases, thus functioning as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be induced by TGF-beta, suggesting its role in TGF-beta induced growth inhibition. In my VEP results, the inversion at chr9:21907128 causes a coding_transcript_variant in CDKN2B.
 3. LINC03142 (long intergenic non-protein coding RNA 3142, GeneID: 139198559) is a non-coding RNA gene located on chromosome 9. According to NCBI, this gene is classified as an ncRNA with no currently established protein product. Its specific function in cancer is not well established. In my VEP results, the deletions at chr9:22999488 and chr9:23362803 cause intron_variant and non_coding_transcript_variants in LINC03142.

- 4. ITPR2 (inositol 1,4,5-trisphosphate receptor type 2, GeneID: 3709) is a protein-coding gene located on chromosome 12. According to NCBI, the protein encoded by this gene belongs to the inositol 1,4,5-trisphosphate receptor family, whose members are second messenger intracellular calcium release channels. These proteins mediate a rise in cytoplasmic calcium in response to receptor activated production of inositol triphosphate, and are involved in many processes including cell migration, cell division, smooth muscle contraction, and neuronal signaling. In VEP results, the deletion at chr12:26358087 causes an intron_variant in ITPR2.
- 5. MTAP (methylthioadenosine phosphorylase, GeneID: 4507) is a protein-coding gene located on chromosome 9p21.3, in close proximity to CDKN2A and CDKN2B. According to NCBI, this gene encodes an enzyme that plays a major role in polyamine metabolism and is important for the salvage pathway of both adenine and methionine. The encoded enzyme is deficient in many cancers. In VEP results, the inversions at chr9:21832461 and chr9:21907128 both cause coding_sequence_variants in MTAP.
- Based on the VEP annotation results and gene information from NCBI, CDKN2A, CDKN2B, and MTAP are the genes most likely to be linked to cancer in the A549 cell line.
- CDKN2A and CDKN2B are both tumor suppressor genes located on chromosome 9p21.3 that control cell cycle G1 progression. Their loss can drive cancer development by allowing uncontrolled cell proliferation. Both genes are affected by the same inversion at chr9:21907128, suggesting a large genomic rearrangement in the 9p21.3 region that simultaneously disrupts both tumor suppressors. Co-deletion of these two genes has been commonly reported in lung adenocarcinoma, which is consistent with A549 being a lung adenocarcinoma cell line.
- MTAP is also located on chromosome 9p21.3 in close proximity to CDKN2A and CDKN2B, and is affected by two separate inversions at chr9:21832461 and chr9:21907128. Since MTAP encodes an enzyme that is deficient in many cancers, its disruption alongside CDKN2A and CDKN2B further supports the involvement of the 9p21.3 region in driving cancer in A549.
- Overall, together with the KRAS G12S mutation identified in Task 2, these structural variants affecting CDKN2A, CDKN2B, and MTAP suggest that the loss of the 9p21.3 region is a key genomic alteration driving cancer in the A549 cell line.